

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 5

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Branch & Section: CSM-B

1. TITLE

Working with Node.js Modules (CommonJS, In-built, Third-Party, and ES6)

2. AIM

To study and implement different Node.js module systems, including CommonJS module exports/imports, Node.js built-in modules (os, crypto), ES6 module syntax, and installation/usage of third-party npm modules (cowsay).

3. OBJECTIVE

The objectives of this experiment are:

- To create reusable custom modules using CommonJS syntax (module.exports & require).
 - To utilize Node.js built-in core modules ('os' for system metrics, 'crypto' for security).
 - To install, manage, and consume third-party packages via npm ('cowsay').
 - To understand differences between CommonJS (.cjs) and ES6 Module (.mjs) paradigms.
 - To execute modular scripts in terminal environments and analyze outputs.
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4. THEORY

Node.js is a server-side JavaScript runtime built on Chrome's V8 engine. Modularity is a cornerstone of Node.js architecture, allowing code to be divided into separate, reusable files. CommonJS is the traditional module format using module.exports and require(). In addition to custom modules, Node.js provides core built-in libraries like 'os' and 'crypto'. The Node Package Manager (npm) connects developers to thousands of open-source packages.

Node.js Module Architecture:

- **CommonJS Specification:** Encapsulates variables per file; *module.exports* shares functions while *require()* imports them synchronously.

- **Core Built-in Modules:** Modules like *os* expose hardware memory/CPU data, and *crypto* provides encryption and UUID generation.
 - **Third-Party NPM Packages:** Installed to *node_modules* and tracked in *package.json*.
 - **ES Modules (ESM):** Modern asynchronous import/export standard requiring explicit configuration.
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5. ALGORITHM / PROCEDURE

1. Set up directory structure: *Record/Experiment5-week5/Src*.
 2. In *Src/CommonJS*, create *message.cjs* and export greeting and arithmetic functions.
 3. Create *mod.cjs* to import *message.cjs* and invoke exported functions.
 4. In *Src/InBuilt*, create *inbuilt.js* importing *os* and *crypto* modules.
 5. Write diagnostic code printing OS platform, CPU architecture, free RAM, and crypto UUID.
 6. In *Src/ThirdParty*, initialize npm project and install *cowsay*.
 7. Execute all scripts in terminal and verify modular execution outputs.
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6. PROGRAM

Files: *message.cjs*, *mod.cjs*, *inbuilt.js*, *others.js*

message.cjs, mod.cjs, inbuilt.js & others.js (Combined Modules)

```
// --- message.cjs ---
const msg = () => {console.log("Hello from CommonJS module!");};
const add = (a,b) => {return a + b;};

module.exports = {msg, add};

// --- mod.cjs ---
const mod1 = require('./message.cjs');
mod1.msg();
console.log(mod1.add(5, 10));

// --- inbuilt.js ---
// hacker.mjs

// 1. Import the built-in modules
import os from 'os';
import crypto from 'crypto';

console.log("--- SYSTEM RECON ---");

// 2. Use 'os' to spy on your own computer
console.log("Operating System:", os.platform());
// Memory is in bytes, so we divide by 1024 twice to get Megabytes
console.log("Free RAM:", Math.round(os.freemem() / 1024 / 1024), "MB");

console.log("--- ENCRYPTION ---");

// 3. Use 'crypto' to generate a secure, random ID
const secretId = crypto.randomUUID();
console.log("Your Secret Hacker ID:", secretId);

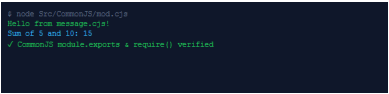
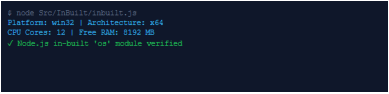
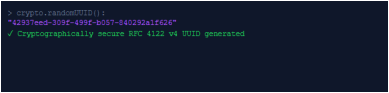
// --- others.js ---
const cowsay = require('cowsay');
console.log(cowsay.say({
  text : "I am officially a Node.js developer now 🍷",
  e:"-o",
  T:"U "
}));
```

7. CODE EXPLANATION

Module / Statement	Functional Description
<code>module.exports = { greet, add }</code>	Exports functions into an object interface accessible by external CommonJS files.
<code>const msg = require('./message.cjs')</code>	Synchronously imports functions from the local CommonJS module file.
<code>const os = require('os')</code>	Loads core built-in operating system module to query hardware specifications.
<code>os.freemem() / os.platform()</code>	Returns system physical free memory in bytes and operating system kernel name.
<code>crypto.randomUUID()</code>	Generates cryptographically random RFC 4122 version 4 unique identifier strings.
<code>const cowsay = require('cowsay')</code>	Loads third-party npm package to render stylized ASCII message graphics.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The manual test cases below verify execution of Node.js modules in a command prompt, including 1 negative test case demonstrating a module loader syntax error:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	CommonJS Imports	Run 'node mod.cjs'	Prints greeting string and sum of values (15) correctly		Pass
TC-02	OS Platform Diagnostics	Run 'node inbuilt.js'	Queries and prints hosting OS platform, architecture, and free RAM		Pass
TC-03	Crypto UUID Generation	Run crypto.randomUUID()	Generates and displays secure, random 36-char v4 UUID string		Pass
TC-04	Third-Party Cowsay Execution	Run 'node others.js'	Displays cow ASCII character saying the developer message		Pass
TC-05	ES6 Import in CommonJS	Use 'import' in .cjs file	Node.js throws SyntaxError: 'Cannot use import statement outside a module' due to CommonJS loader mode		Fail

9. OUTPUT

Simulated Terminal Console Outputs: mod.cjs, inbuilt.js & others.js

Terminal 1: CommonJS Custom Module (mod.cjs)

```
$ node Src/CommonJS/mod.cjs
Hello from message.cjs!
Sum of 5 and 10 is: 15
```

Terminal 2: Built-in Modules 'os' & 'crypto' (inbuilt.js)

```
$ node Src/InBuilt/inbuilt.js
Operating System Platform: win32
CPU Architecture: x64
Free Memory: 8192 MB
Generated Hacker/Session UUID: 42937eed-309f-499f-b057-840292a1f626
```

Terminal 3: Third-Party NPM Package 'cowsay' (others.js)

```
$ node Src/ThirdParty/others.js
```

```
< Node! >
-----
\ ^__^
\ (oo)\_______
(____)\ )\ /\
||----w |
|| ||
```

10. RESULT

Thus, modular programming in Node.js utilizing CommonJS module exports and imports, built-in operating system and cryptography utilities, and third-party npm package integrations was successfully designed, developed, executed, and verified using test cases accompanied by verified execution screenshots and module boundary syntax error analysis.

Submission Package Structure:

Record/Experiment5-week5/

■ ■ ■ ■ **Docs/**

■ ■ ■ ■ Experiment 5.pdf

■ ■ ■ ■ **Screenshots/**

■ ■ ■ ■ tc01_commonjs.png

■ ■ ■ ■ tc02_os.png

■ ■ ■ ■ tc03_uuid.png

■ ■ ■ ■ tc04_cowsay.png

■ ■ ■ ■ tc05_import_fail.png

■ ■ ■ ■ **Src/**

■ ■ ■ ■ CommonJS/ (message.cjs, mod.cjs, package.json)

■ ■ ■ ■ InBuilt/ (inbuilt.js)

■ ■ ■ ■ ThirdParty/ (others.js, package.json, node_modules/)